

CHE 348 NUMERICAL METHODS IN CHE/PROB SOLV: Spring 2023.
(Unique Number: 15585)

Course Description: Numerical solutions to algebraic and differential equations; numerical methods to integration, interpolation, and regression analysis, with application to chemical engineering. Two lectures (75 minutes each) and one recitation hour per week.

Prerequisites: ChE 210 (Introduction to Computing), ChE 317 (Introduction to Chemical Engineering Analysis), and M 427K (Differential Equations) with a grade of at least C- in each.

This class assumes that you have a good grasp of the following subjects:

- Algebra and mathematical (trigonometric/exponential/logarithmic) functions;
- Integration and differentiation;
- Matrices and (ordinary/partial) differential equations and their solutions;
- Physical and chemical behaviors (e.g., ideal/real gas behavior, mass and energy conservation)

The objectives of this course:

- To develop the confidence necessary to successfully solve mathematical problems with a computer;
- To be able to formulate and write structured computer code;
- To understand the formulation behind basic numerical methods for matrix manipulation, finding roots, rudimentary optimization and numerical integration of ordinary and partial differential equations; and
- To be able to solve linear and non-linear algebraic equations, and to integrate coupled sets of most ordinary and partial differential equations.

Instructor: Professor Jim Chelikowsky

Office: POB 4.324 Office Hours: Zoom appointments or face to face meetings as requested.
Phone: (512) 232-9083. Email: jrc@utexas.edu, please put "348" in the subject header.

Teaching Assistant (TA) : Adhika Retnanto, Email: aretnanto@utexas.edu. Adhika will be happy to answer email questions and setup meetings as requested

Class Meeting Times: Lectures TuTh 12:30-2:00 p.m. CPE 2.218. Recitation Sections:
F 11-12:00 a.m. CPE 2.218

Recitations: The teaching assistant will conduct recitations where Matlab coding will be reviewed and/or a problem similar to one of those on the CEs will be solved. This is an excellent opportunity to get advice while you are working on a computer. Please go to recitation and take advantage of it.

Web Site: http://tesla.oden.utexas.edu/chen_348_2023

The website has "everything" from lecture notes to book chapters to old exams. Password provided in class.

Text: *Applied Numerical Methods with MATLAB*, S.C. Chapra, McGraw Hill, 4th edition, or early editions. (Strongly recommended, but not required).

Evaluation: We will distribute comprehensive evaluations (CEs) on a regular basis. You must work on these on your own. You may consult your notes, book and even the internet – but you may not copy and paste code from anybody or any resource. There will be 4 or 5 CEs, this will account for 80% of your grade. About every two or three weeks, there will be an online quiz. These quizzes will be short, e.g., 10-15 multiple choice questions and can be done at your convenience within a given time window (usually a day or so). They will count for 20% of your grade. There will be no makeup quizzes. Legitimately missed quiz scores will be replaced by your average over the remaining quizzes.

Classroom Accessibility and Inclusivity: The university is committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, and need accommodations please contact Services for Students with Disabilities (SSD). Please refer to SSD's website for contact and more information: <http://diversity.utexas.edu/disability/>. If you are already registered with SSD, please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations and needs in this course.

Title IX Reporting: Title IX is a federal law that protects against sex and gender-based discrimination, sexual harassment, sexual assault, sexual misconduct, dating/domestic violence and stalking at federally funded educational institutions. When sexual misconduct occurs in our community, the university can: (1) Intervene to prevent harmful behavior from continuing or escalating. (2) Provide support and remedies to students and employees who have experienced harm or have become involved in a Title IX investigation. (3) Investigate and discipline violations of the university's relevant policies.

Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) *must be reported*. If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email advocate@austin.utexas.edu. For more information about reporting options and resources, visit <http://www.titleix.utexas.edu/>, contact the Title IX Office via email at titleix@austin.utexas.edu, or call 512-471-0419.

COVID-19 Update: You are encouraged to stay up-to-date on the latest news as related to the student experience. <https://protect.utexas.edu/>

Tentative class schedule: Subject to change

Week of	Topics to be covered
January 9	Introduction to MatLab
January 16	Solution of a single nonlinear algebraic equation by bracketing methods (Bisection).
January 23	Solution of a single nonlinear algebraic equation by open methods (Newton Raphson, Secant methods).
January 30	Matrices: Their representation, special matrices, algebra of matrices.
February 6	Solution of sets of linear algebraic equations by Gaussian Elimination with and without pivoting.
February 13	LU decomposition
February 20	Matrix inversion
February 27	Solution of multiple sets of nonlinear equations by generalized Newton method
March 6	Finite difference techniques for numerical differentiation. Trapezoidal and Simpson's rule for numerical integration. Monte Carlo methods.
March 13	Spring break
March 20	Explicit and implicit methods for ordinary differential equations
March 27	Predictor-Corrector methods for ordinary differential equations
April 3	Shooting method for boundary value problems.
April 10	Finite difference method for boundary value problems
April 17	Curve fitting and interpolation, splines